Human Activity Recognition with Missing-Joint Recovery

A measured extension of a published skeleton-action model. We investigate whether limited repair of temporarily missing pose joints helps an unchanged pretrained classifier, and whether calibrated rejection makes accepted predictions more reliable.

Base paper and verified journal status

Continual Spatio-Temporal Graph Convolutional Networks
Lukas Hedegaard, Negar Heidari and Alexandros Iosifidis.
Pattern Recognition 140 (2023), 109528. Published paper.

Q1 evidence: SCImago/SJR **2024**, category **Computer Vision and Pattern Recognition**; journal SJR 2.058, Q1. This verifies that named ranking/year, not 2023 JCR status. SCImago category listing.

The implementation in one pipeline

Real OpenPose skeleton sequence > optional bounded causal repair > authors' fixed CoST-GCN* classifier > optional calibrated rejection.

The model combines information across body joints and across time. The paper's continual formulation reuses intermediate calculations during streaming. Our measurements use its weight-compatible clip path, with a recorded numerical check against the continual definition.

What is ours

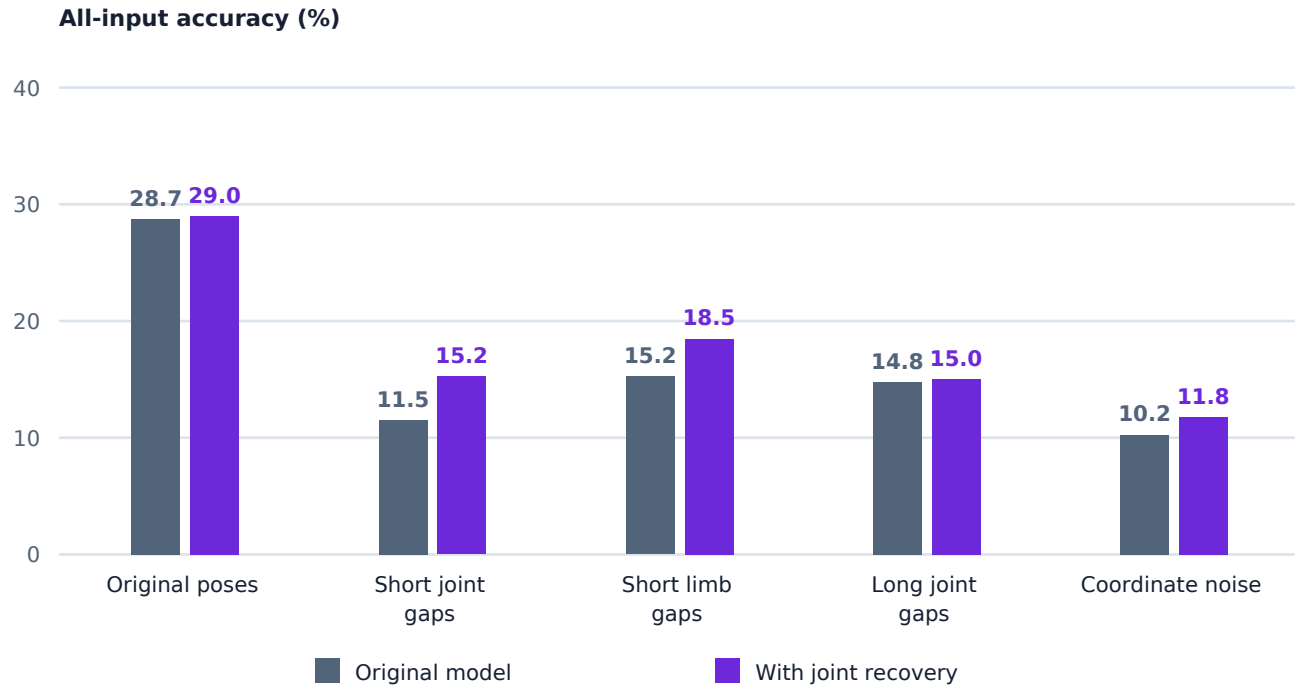
We implement pose repair, confidence decay, bounded gap handling, rejection calibration, controlled corruption experiments, ablations and a reviewable comparison. The pretrained classifier and its weights are credited to the authors; they were not trained by this project.

Frozen setting	Measured configuration
Calibration-selected recipe	confident_hold_5
Repair behavior	Confidence-gated hold (beta = 0)
Gap and update parameters	Maximum 5 frames; alpha 1; beta 0
Prior confidence / decay	Minimum repair confidence 0.5; decay 0.9
Coordinate guards	Speed <= 0.08; displacement <= 0.2 (input units)
Person-slot reset guard	Torso jump > 0.15 input units resets prior state

Measured run: 2026-10-11T23:17:40.737167+05:30. 200 calibration clips; 400 disjoint test clips; 400 classifier outputs.

Recovery gains and individual regressions

The primary contrast isolates recovery without rejection. Both methods see the same held-out clips, the same preprocessing and the same added corruption masks. All five conditions are retained.



Condition	Difference	Paired 95% interval	Fixed / regressed
Original poses	+0.25 pp	-1.50 pp to +2.00 pp	7 / 6
Short joint gaps	+3.75 pp	+1.50 pp to +6.25 pp	21 / 6
Short limb gaps	+3.25 pp	+0.75 pp to +5.75 pp	20 / 7
Long joint gaps	+0.25 pp	-1.50 pp to +2.00 pp	7 / 6
Coordinate noise	+1.50 pp	+0.00 pp to +3.00 pp	8 / 2

Differences are percentage points, not relative percentage improvements. A paired bootstrap resamples clip pairs; intervals crossing zero do not establish a consistent gain in this pilot. Corrected and regressed clips are both shown.

Noise stress test: recovery changes accuracy by +1.50 pp. The paired interval includes zero, so a consistent improvement is not established here. Gap repair is not a general-purpose denoising solution.

Read accuracy alongside coverage

All-input accuracy and macro-F1 count abstentions as errors. Accepted accuracy uses only clips receiving a prediction. A higher accepted accuracy at lower coverage is a tradeoff, not proof of a better classifier.

Condition	Variant	All-input accuracy	Macro-F1	Accepted accuracy	Coverage
Original poses	Original model	28.75%	27.33%	28.75%	100.00%
	Joint recovery	29.00%	27.98%	29.00%	100.00%
	Rejection only	28.50%	27.98%	33.93%	84.00%
	Recovery + rejection	28.75%	28.42%	33.82%	85.00%
Short joint gaps	Original model	11.50%	8.78%	11.50%	100.00%
	Joint recovery	15.25%	12.66%	15.25%	100.00%
	Rejection only	11.50%	9.15%	15.03%	76.50%
	Recovery + rejection	15.00%	12.64%	19.11%	78.50%
Short limb gaps	Original model	15.25%	13.43%	15.25%	100.00%
	Joint recovery	18.50%	16.34%	18.50%	100.00%
	Rejection only	14.50%	13.31%	18.18%	79.75%
	Recovery + rejection	17.75%	16.44%	21.98%	80.75%
Long joint gaps	Original model	14.75%	13.21%	14.75%	100.00%
	Joint recovery	15.00%	12.95%	15.00%	100.00%
	Rejection only	14.50%	13.09%	18.53%	78.25%
	Recovery + rejection	14.75%	13.12%	18.44%	80.00%
Coordinate noise	Original model	10.25%	8.92%	10.25%	100.00%
	Joint recovery	11.75%	10.49%	11.75%	100.00%
	Rejection only	10.00%	8.88%	11.53%	86.75%
	Recovery + rejection	10.75%	9.47%	12.57%	85.50%

Macro-F1 averages the 251 true classes represented in the test subset. The model retains all 400 outputs. 2 empty test clips are retained.

Frozen rejection thresholds: original inputs 0.080; repaired inputs 0.080. Threshold selection seeks the highest calibration accepted accuracy subject to at least 80% calibration coverage; this does not guarantee that coverage on unseen conditions.

What this experiment establishes

- Original poses: accuracy difference +0.25 pp; interval includes zero.
- Short joint gaps: accuracy difference +3.75 pp; paired interval excludes zero.
- Short limb gaps: accuracy difference +3.25 pp; paired interval excludes zero.
- Long joint gaps: accuracy difference +0.25 pp; interval includes zero.
- Coordinate noise: accuracy difference +1.50 pp; interval includes zero.

Measured processing cost

Condition	Original ms/clip	Recovery ms/clip
Original poses	31.91	47.15
Short joint gaps	30.87	43.58
Short limb gaps	31.04	43.68
Long joint gaps	31.06	43.49
Coordinate noise	30.94	43.79

Mean synchronized model processing time per 300-frame clip in batches; recovery runs on CPU. Excludes transfers, camera pose estimation, I/O and browser. Not per-frame streaming latency.

Limits that must stay in the presentation

- Pose coordinates and labels are real Kinetics data. Joint masking and coordinate noise are synthetic perturbations; physical camera occlusion has not been validated.
- This is a sampled pilot, not a reproduction of the full paper benchmark or a state-of-the-art claim. Upstream validation data may have influenced the authors' checkpoint selection.
- Confidence-ranked person slots may change identities. Long gaps, abrupt motion and person swaps remain difficult. The optional identity guard is not a full tracker.
- The new 400-class classifier has no validated webcam integration. The existing MediaPipe six-activity dashboard is a separate demo, not this research checkpoint.

A result you can inspect and explain

Reproduction

Clone the project repository and install its recorded research dependencies. Setup downloads a pinned upstream checkout and verifies the checkpoint hash. The data utility downloads a fixed real-data subset using byte ranges rather than the full archive.

```
python -m pip install -r requirements-research.txt
python scripts/setup_research.py
python scripts/research_data.py
python -m research.benchmark --phase all
python scripts/render_research_report.py
```

The Colab notebook is notebooks/Research_Comparison.ipynb. Keep comparison.json, predictions.csv, calibration.json, design.json and the data manifest together. Selection is deterministic; never choose test clips based on whether predictions succeed.

Short viva answers

Why a graph network? A body is naturally a graph of connected joints. Spatial layers combine neighboring joints; temporal layers learn movement across frames.

What is the contribution? A bounded repair and rejection pipeline plus a controlled, reproducible ablation study around the authors' unchanged pretrained classifier.

Why is repair causal? It uses previous and current measurements only. Filling a gap using a future observation would violate the streaming constraint.

Why not compare our number directly with the paper? The paper uses full benchmark splits. This smaller pilot and its added corruptions answer a different, explicitly scoped question.

Why show negative results? They identify the method's limits. A useful experiment must show regressions and unchanged conditions as well as improvements.

Provenance and evidence

Numerical clip/continual check: passed. Maximum logit differences: clip vs continual 2.86e-06; clip vs step 4.77e-06. This is a check on the recorded calibration input, not validation of every possible stream.

Published base paper | Authors' implementation | SCImago 2024 category evidence

Upstream commit: a8fe2937a33f24cce65c1f8c2fc41081bceda721
Checkpoint SHA256: bec5c9563cc017a6e4739c2255ad76b24a046e445210b8fc965816435229452f
Comparison JSON SHA256: 346808497b94c5ee389f38629a4e46429b4895611430b73fae048e29643986b9